

LEEMING, United Kingdom

WMO: 032570

Lat: 54.2969N

Lon: 1.5331W

Elev: 40

StdP: 100.84

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.5	(c) -2.8	(d) -6.2	(e) 2.2	(f) -2.4	(g) -4.9	(h) 2.5	(i) -0.9	(j) 16.4	(k) 9.7	(l) 14.5	(m) 8.3	(n) 1.3	(o) 160	(p) 0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 24.9	(d) 17.9	(e) 23.0	(f) 17.0	(g) 21.3	(h) 16.2	(i) 18.7	(j) 23.3	(k) 17.8	(l) 21.8	(m) 16.9	(n) 20.3	(o) 4.3	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.2	20.6	16.1	11.5	19.6	15.3	10.9	18.6	53.2	23.3	50.3	21.9	47.5	20.3	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.2	10.5	9.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-7.9	28.3	2.2	1.6	-9.5	29.4	-10.8	30.3	-12.0	31.2	-13.6	32.3		
			-8.2	20.6	2.2	1.2	-9.8	21.5	-11.1	22.2	-12.4	22.8	-14.0	23.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	4.1	4.5	6.0	8.1	10.9	13.8	16.1	15.7	13.4	10.2	6.6	4.2
	DBStd	5.20	3.17	3.21	2.92	2.89	2.87	2.56	2.38	2.38	2.35	2.92	3.22	3.75
	HDD10.0	884	182	154	127	71	23	2	0	0	2	33	109	181
	HDD18.3	3239	440	387	382	308	229	137	78	87	149	251	353	437
	CDD10.0	704	1	1	3	13	52	117	188	178	103	40	6	2
	CDD18.3	17	0	0	0	0	0	2	8	7	1	0	0	0
	CDH23.3	137	0	0	0	0	3	18	63	47	5	1	0	0
	CDH26.7	16	0	0	0	0	0	2	7	8	0	0	0	0
(14) Wind	WSAvg	4.1	4.8	4.8	4.6	4.1	4.1	3.8	3.5	3.6	3.7	4.0	4.2	4.3
(15) Precipitation	PrecAvg	628	56	42	43	46	48	57	53	64	53	59	59	56
	PrecMax	927	125	108	108	137	131	178	139	165	174	134	152	129
	PrecMin	442	7	3	8	2	5	9	9	14	5	14	16	16
	PrecStd	85	27	23	22	31	27	39	30	31	35	29	29	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.4	15.7	20.1	23.3	26.0	27.6	27.8	23.9	19.1	15.2	13.8
		MCWB	10.7	11.0	11.0	13.3	16.3	17.9	18.7	18.7	17.7	15.6	13.0	11.8
	2%	DB	11.3	11.8	13.6	16.8	20.6	22.7	25.1	24.2	21.1	17.1	13.9	12.3
		MCWB	9.4	9.6	10.1	11.8	14.9	16.5	18.1	17.9	16.1	14.2	11.9	10.6
	5%	DB	10.1	10.7	12.2	14.9	18.4	20.8	23.2	21.9	19.2	15.9	12.7	11.1
		MCWB	8.3	8.5	9.3	10.7	13.4	15.4	17.2	16.9	15.0	13.5	10.8	9.4
	10%	DB	9.0	9.6	11.0	13.4	16.5	19.1	21.2	20.3	17.8	14.8	11.5	9.8
		MCWB	7.4	7.5	8.3	9.9	12.2	14.5	16.3	15.9	14.2	12.5	9.9	8.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	11.0	11.7	14.0	17.2	19.5	20.0	19.9	18.5	16.3	13.6	12.0
		MCDB	12.4	12.9	14.3	17.8	22.3	24.2	25.4	25.1	22.4	18.4	15.0	13.4
	2%	WB	9.6	9.9	10.6	12.4	15.4	17.2	18.8	18.6	17.0	14.9	12.2	10.8
		MCDB	11.2	11.7	13.1	15.9	19.8	21.2	23.4	22.9	20.3	16.6	13.5	12.1
	5%	WB	8.5	8.8	9.6	11.2	14.0	16.0	17.9	17.6	15.7	13.9	11.1	9.6
		MCDB	9.9	10.5	11.8	14.2	17.5	19.7	22.0	21.2	18.1	15.5	12.4	10.9
	10%	WB	7.4	7.7	8.6	10.2	12.7	15.0	16.8	16.7	14.7	12.8	10.0	8.3
		MCDB	8.8	9.4	10.8	13.1	15.8	18.4	20.5	19.5	17.0	14.4	11.3	9.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.1	7.3	8.5	8.9	8.5	8.8	8.6	8.2	6.9	5.9	5.4
		MCDBR	6.1	6.6	9.1	11.0	12.2	12.0	12.2	10.9	10.5	7.5	6.4	6.2
	5% WB	MCWBR	5.6	5.3	6.3	6.8	7.3	6.7	6.1	5.6	6.3	5.4	5.5	5.5
		MCWBR	6.0	6.4	8.5	10.2	11.3	10.7	10.8	9.8	9.0	7.2	6.1	6.1
(40) Clear-Sky Solar Irradiance	taub		0.322	0.337	0.368	0.395	0.395	0.397	0.397	0.385	0.365	0.347	0.321	0.306
		taud	2.305	2.359	2.280	2.228	2.275	2.294	2.299	2.349	2.411	2.425	2.369	2.284
	Ebn at Noon		619	748	800	828	850	851	842	831	802	728	620	551
		Edh at Noon	58	77	102	123	124	123	120	108	90	71	54	49
(44) All-Sky Solar Radiation	RadAvg		0.61	1.25	2.33	3.69	4.71	4.80	4.63	3.82	2.77	1.54	0.76	0.47
	RadStd		0.07	0.14	0.30	0.40	0.27	0.42	0.42	0.25	0.25	0.13	0.07	0.05

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page